

House Price Prediction

Linear Regression Mini Project

Dataset: California Housing

Objective

- Predict house prices using housing features.
- Train a Linear Regression model.
- Evaluate using MAE, RMSE and R^2 score.

Workflow

1. Load dataset
2. Perform EDA
3. Select features and target
4. Split data (80/20)
5. Train model
6. Predict
7. Evaluate

EDA Findings

- Dataset inspected using `head()`, `info()`, `describe()`.
- Missing values checked.
- Histograms plotted.
- Correlation heatmap analyzed.

Model & Evaluation

- Algorithm: Linear Regression
- Metrics: MAE, RMSE, R^2 Score
- Actual vs Predicted scatter plot
- Residual plot

Conclusion

- Linear Regression establishes a baseline model.
- Model performance depends on feature relationships.
- Future work: Random Forest, XGBoost, feature engineering.